OpenBioWorkstation

A workstation for agent-assisted bioinformatics

On-demand tool adaptation and inspectable workflow execution

Visual workflows • Snakemake execution • MCP access • Local run records

Researcher, agent and workstation

Researcher

Define the question and methods

Review parameters and interpretation

Agent / MCP

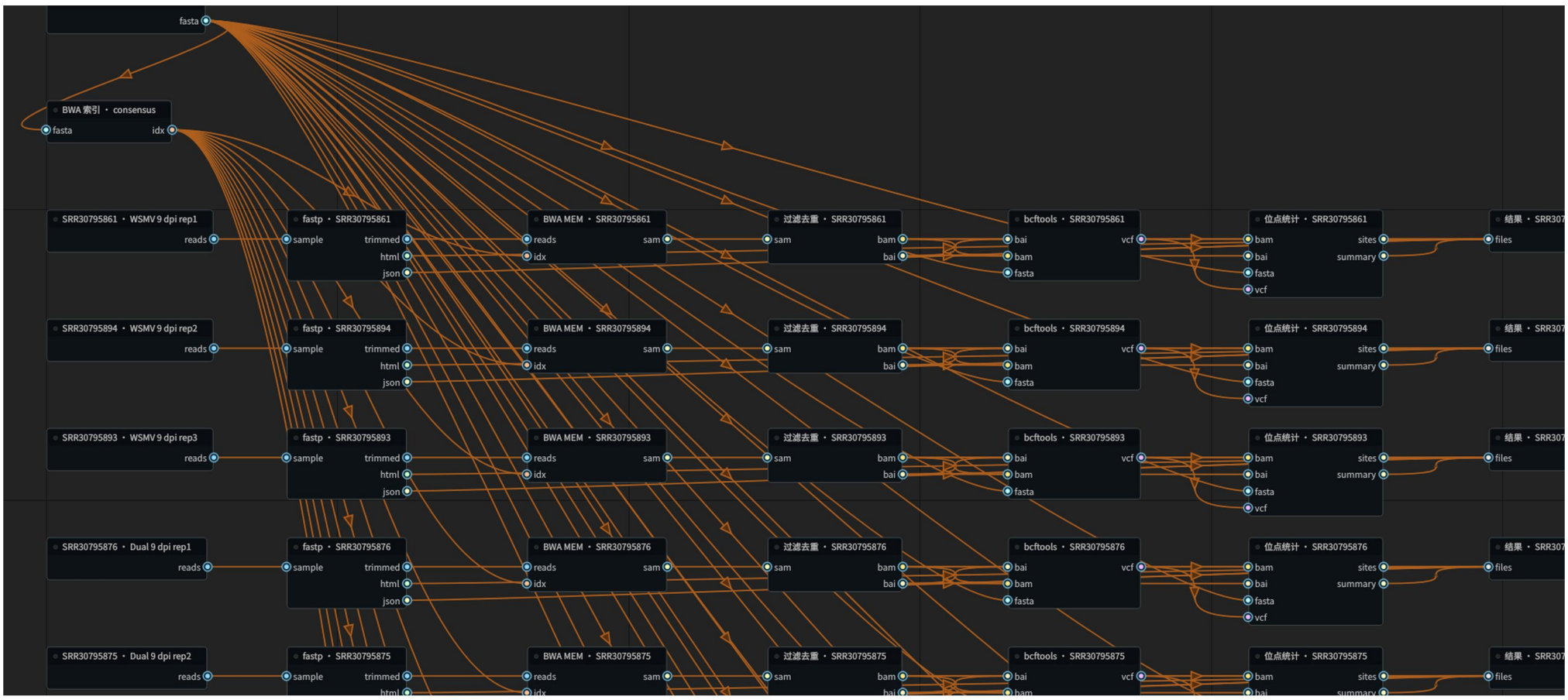
Discover tools and build workflows

Start runs and inspect outputs

Workstation / Snakemake

Execute the analysis workflow

Retain graphs, logs and output files



WSMV workflow excerpt: sample chains share a reference and index. Case-specific nodes are included.

Port format checks, execution states, partial reruns and file previews.

Tool integration and discovery

Local snapshot / 2026-11-01

69

software packages adapted locally

15,524

searchable catalog entries

78 adapted nodes + 5 built-in wrappers + 10 local nodes = 93 node definitions

62 nodes carry review flags; tool review is progressing incrementally.

Packages, nodes and catalog entries are distinct counts. Indexed entries are not necessarily executable.

On-demand tool adaptation with an LLM

Extract CLI help — Prepare the environment → **Generate a definition** — Describe ports and commands → **Check and trial** — Test with synthetic inputs → **Review before use** — Inspect and mark reviewed

Configure the model endpoint and dependencies. Generation and synthetic trials do not validate real-data analysis.

Local adaptations are not installed by default. Check compatibility with each end-user agent client.

Application case: WSMV / TriMV coinfection

12 RNA-seq libraries; no tested site (0/73) reached FDR < 0.10. The result does not establish equivalence.

Source code, agent tutorial and research examples

github.com/ArchitektSpeer/openbioworkstation

Linux • GNU GPL v3.0